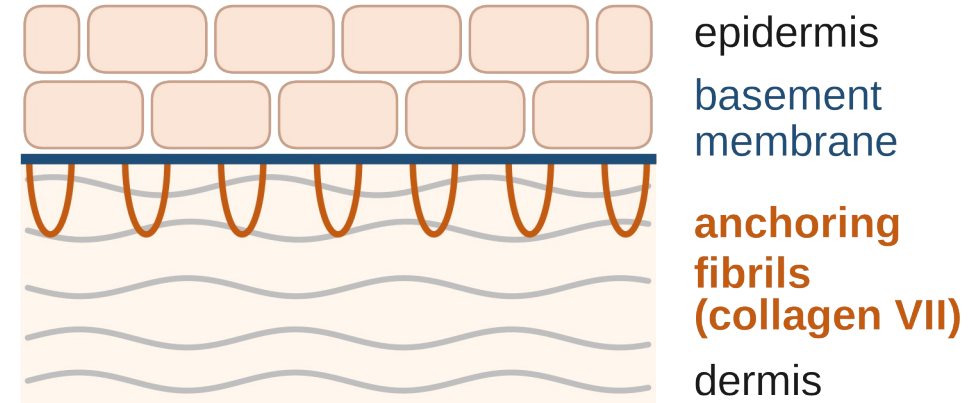


Exercise A.2 – Collagen VII

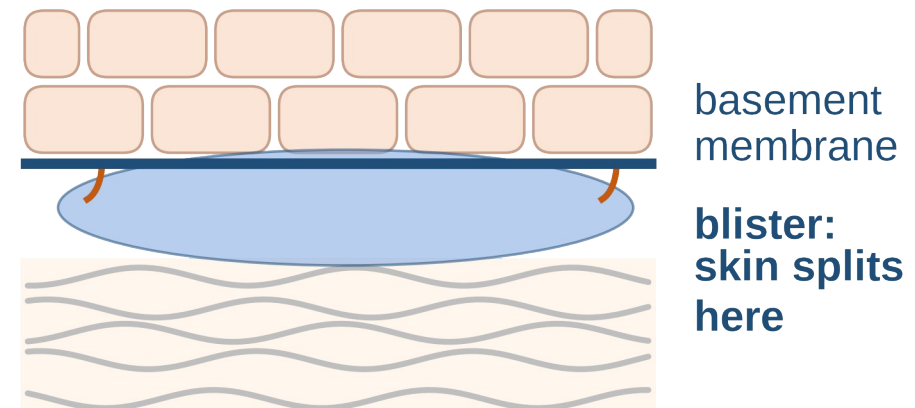
Collagen VII anchors the epidermis

- Collagen VII forms **anchoring fibrils** under the basement membrane
- They tie the epidermis to the dermis
- **Dystrophic epidermolysis bullosa (DEB):** mutations in *COL7A1*
- The skin splits below the basement membrane and blisters after mild friction

Healthy skin

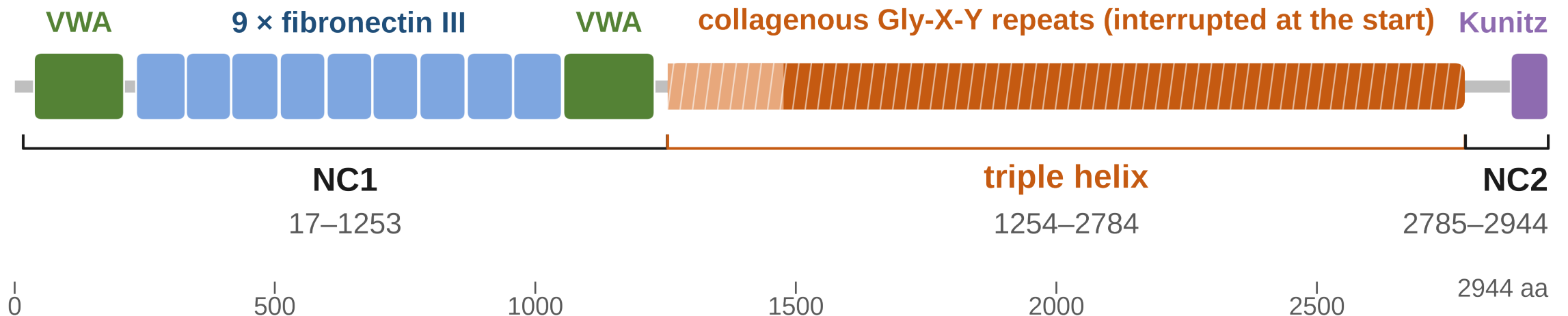


DEB



COL7A1: a protein of 2944 amino acids

- NC1 binds the basement membrane; the long middle part is a triple helix
- Look for this structure in the dotplot of the exercise



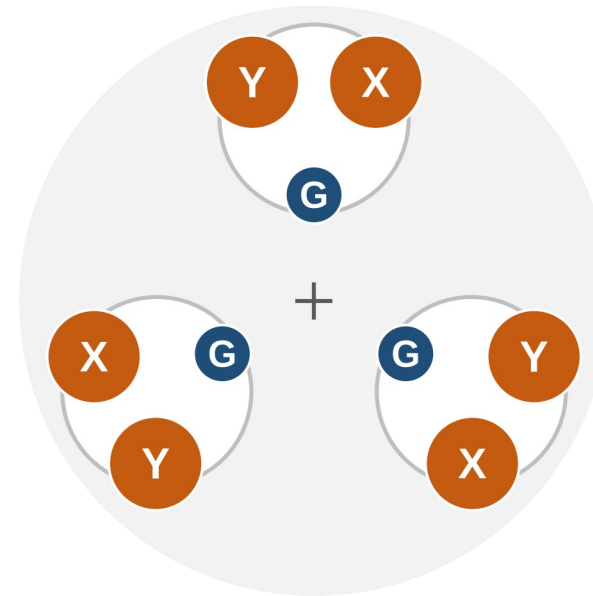
Gly-X-Y: glycine at every third position

- Three chains coil into a triple helix; only glycine fits into its crowded centre
- X and Y are often proline and hydroxyproline

COL7A1, residues 2401–2430

G VV	G FP	G QT	G PR	G EM
G QP	G PS	G ER	G LA	G PP

triple helix, cross-section



Gly points to
the axis:
no room for
a side chain

X, Y on
the surface

G X Y ← one triplet

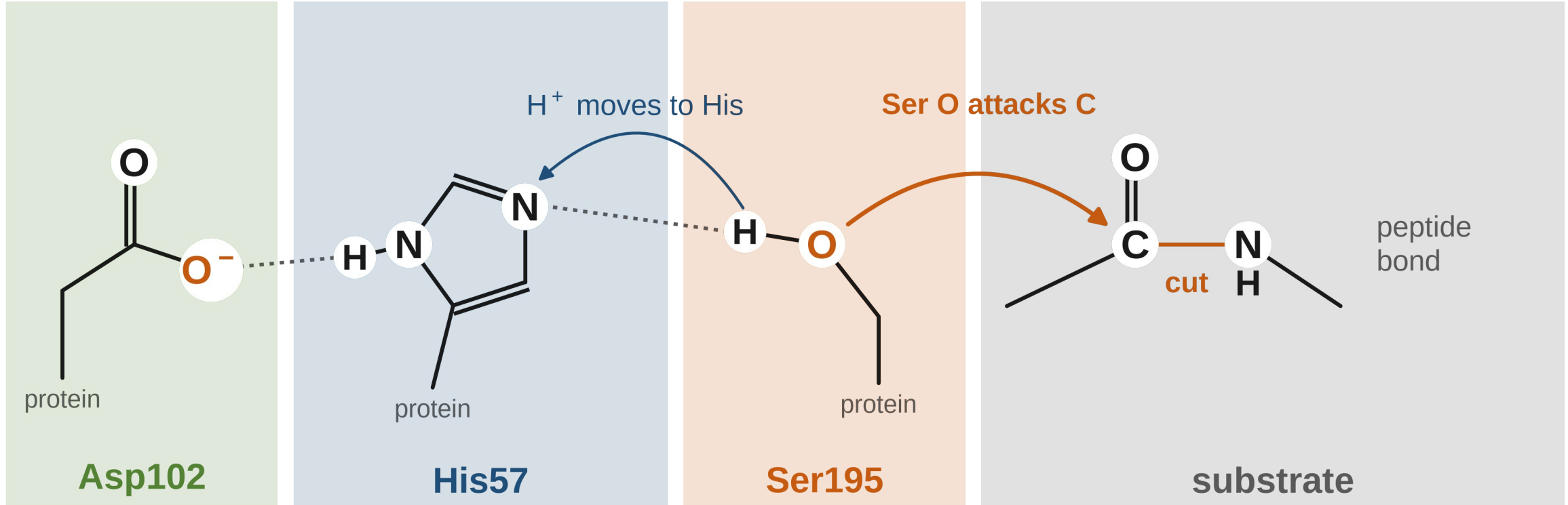
G = glycine (side chain: H)

P = proline

Exercise 3.6 – Serine proteases

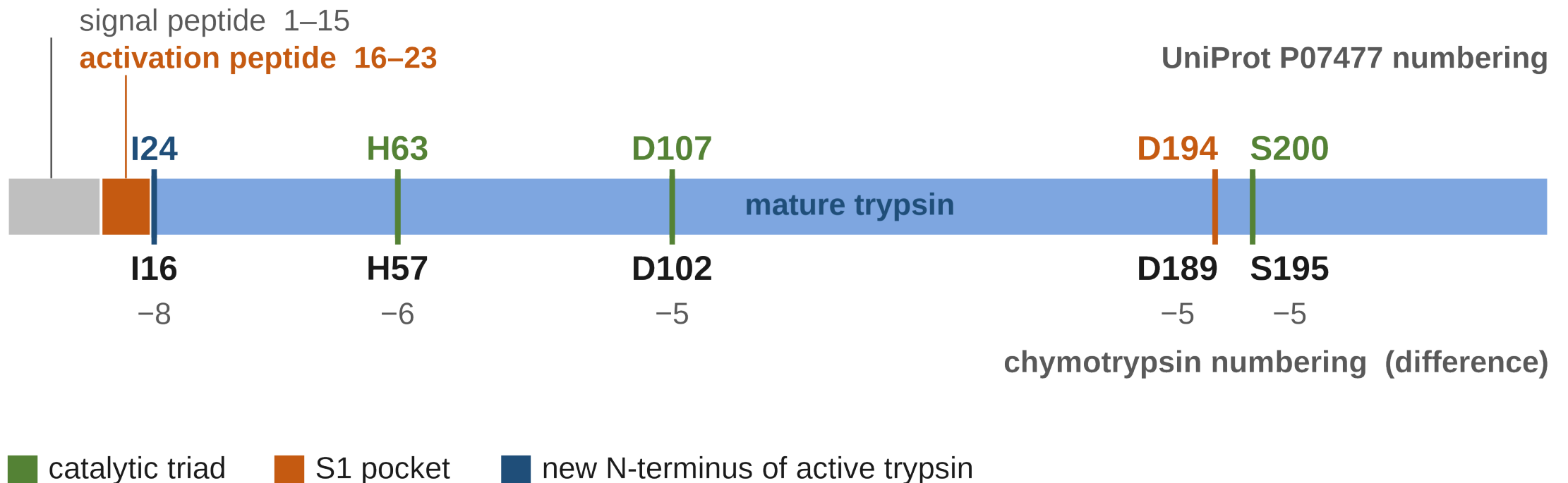
One mechanism, many jobs

- Digestion, blood clotting, immune defence, insect development
- Catalytic triad: Asp and His activate Ser, which attacks the peptide bond



Why His57 is His63 in UniProt

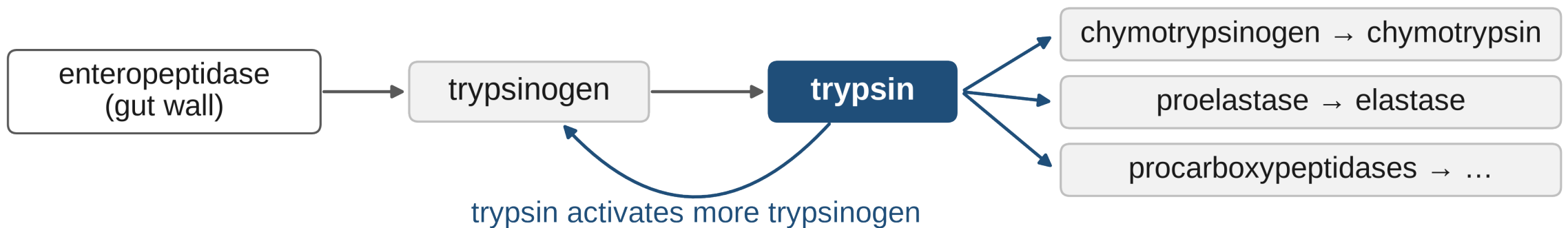
- Papers use chymotrypsin numbering (bovine chymotrypsinogen)
- UniProt counts from Met1 of the precursor of each protein



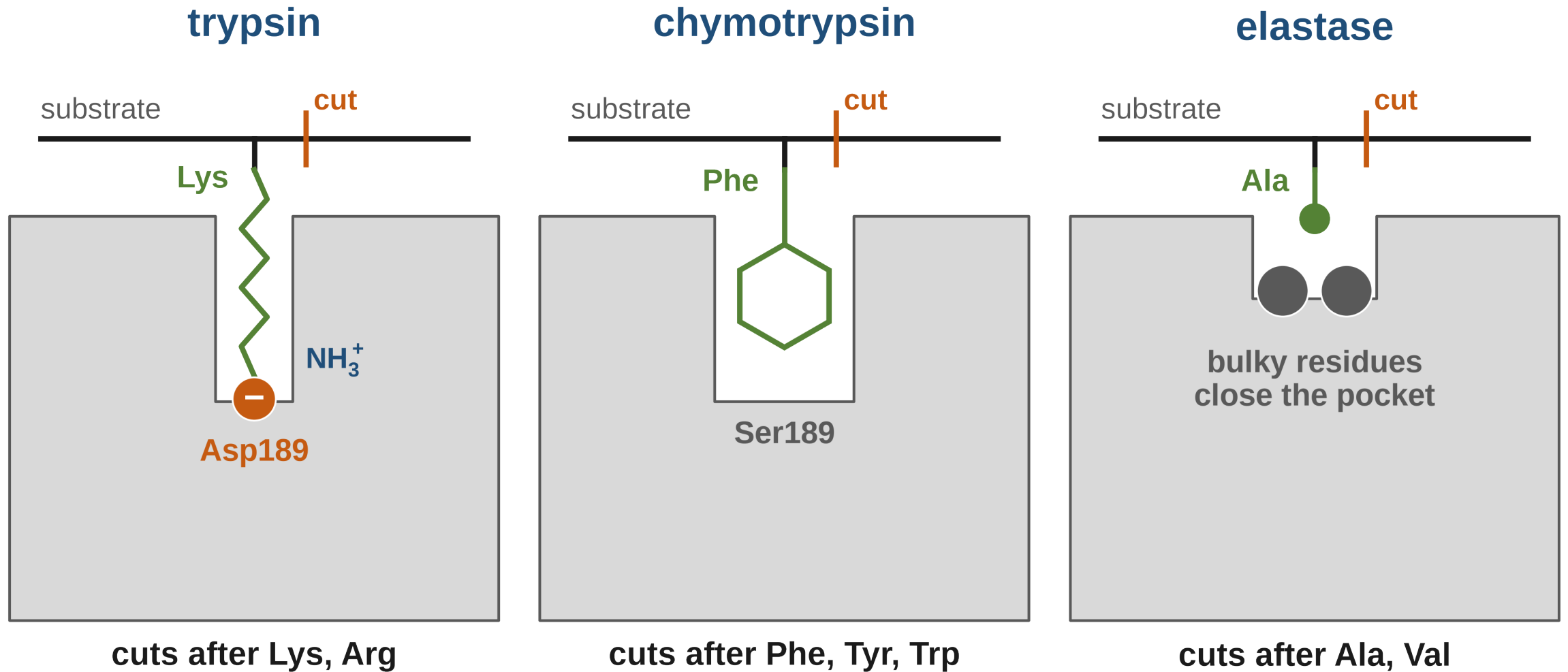
Zymogens: made inactive, switched on by a cut

- Enteropeptidase in the gut cuts trypsinogen after **DDDDK**
- Trypsin then activates the other pancreatic zymogens

enteropeptidase cuts here
... **APFDDDDK** | **IVGG**YNCE ... the new end folds into the protein: active site ready
activation peptide (16–23) new N-terminus IVGG



The S1 pocket decides where the enzyme cuts



chymotrypsin numbering; schematic, not to scale

Exercise 3.8 – Primers for dodder

Dodder: a plant that feeds on other plants

- No roots, little chlorophyll
- **Hauatoria** grow into the host and take water and sugars
- *C. campestris* is a weed of alfalfa, tomato and sugar beet
- About 200 species, hard to tell apart by their flowers



Rickjpelleg, CC BY 2.5

a yellow mat
over the host
plants

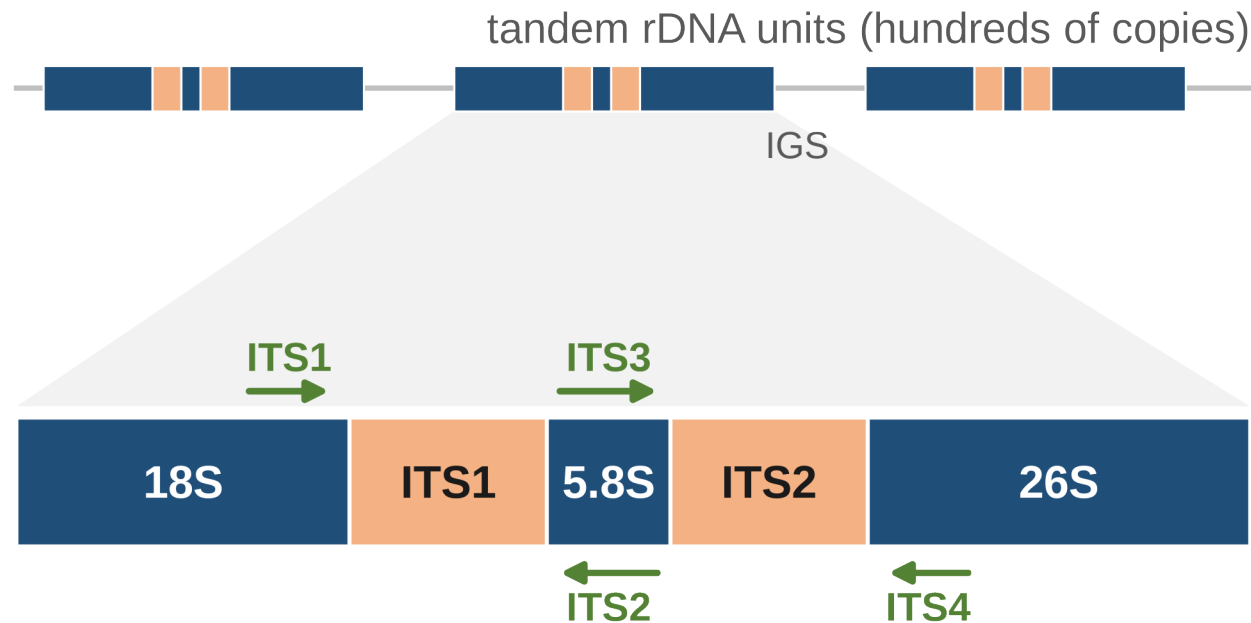


David Eickhoff, CC BY 2.0

leafless stems
coiled around
a host

rDNA: conserved genes, variable spacers

- Primers sit in the conserved genes, the variable spacers tell species apart



toy alignment

sp. A	TCCGTAGG	CCGTCTTTAGCT	GAACGCAG
sp. B	TCCGTAGG	TGGTTGC - A - CT	GAACGCAG
sp. C	TCCGTAGG	TAG - GTTACCGG	GAACGCAG
sp. D	TCCGTAGG	TGGTGA - ACAAT	GAACGCAG
sp. E	TCCGTAGG	TGTTGTCC - CCG	GAACGCAG

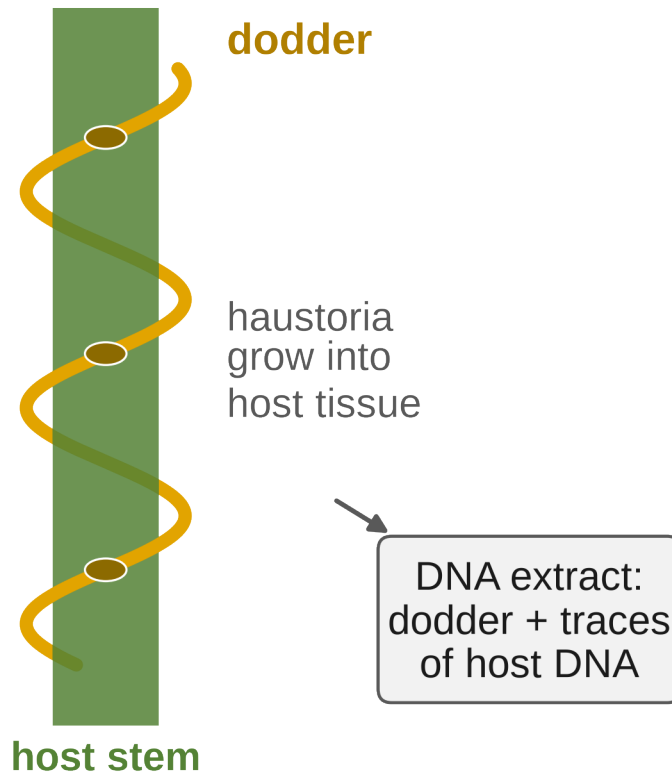
Below the alignment, green arrows point to the first and last columns, labeled 'primer'. A red arrow points to the middle columns, labeled 'species signal'.

genes: conserved **spacers: variable**

universal ITS primers (White et al. 1990); not to scale

The sample can be contaminated with host DNA

- Universal primers amplify the host as well
- A mismatch at the primer's 3' end stops the polymerase



dodder template: perfect match

primer 5' **GACTCGTAGCTTGCA**GTTCC → **new DNA: extended**
template 3' CTGAGCATCGAACGTCAAGGT

host template: mismatch at the 3' end

primer 5' **GACTCGTAGCTTGCA** 3' **no extension**
template 3' CTGAGCATCGAACG^XCAAGGT

toy sequences